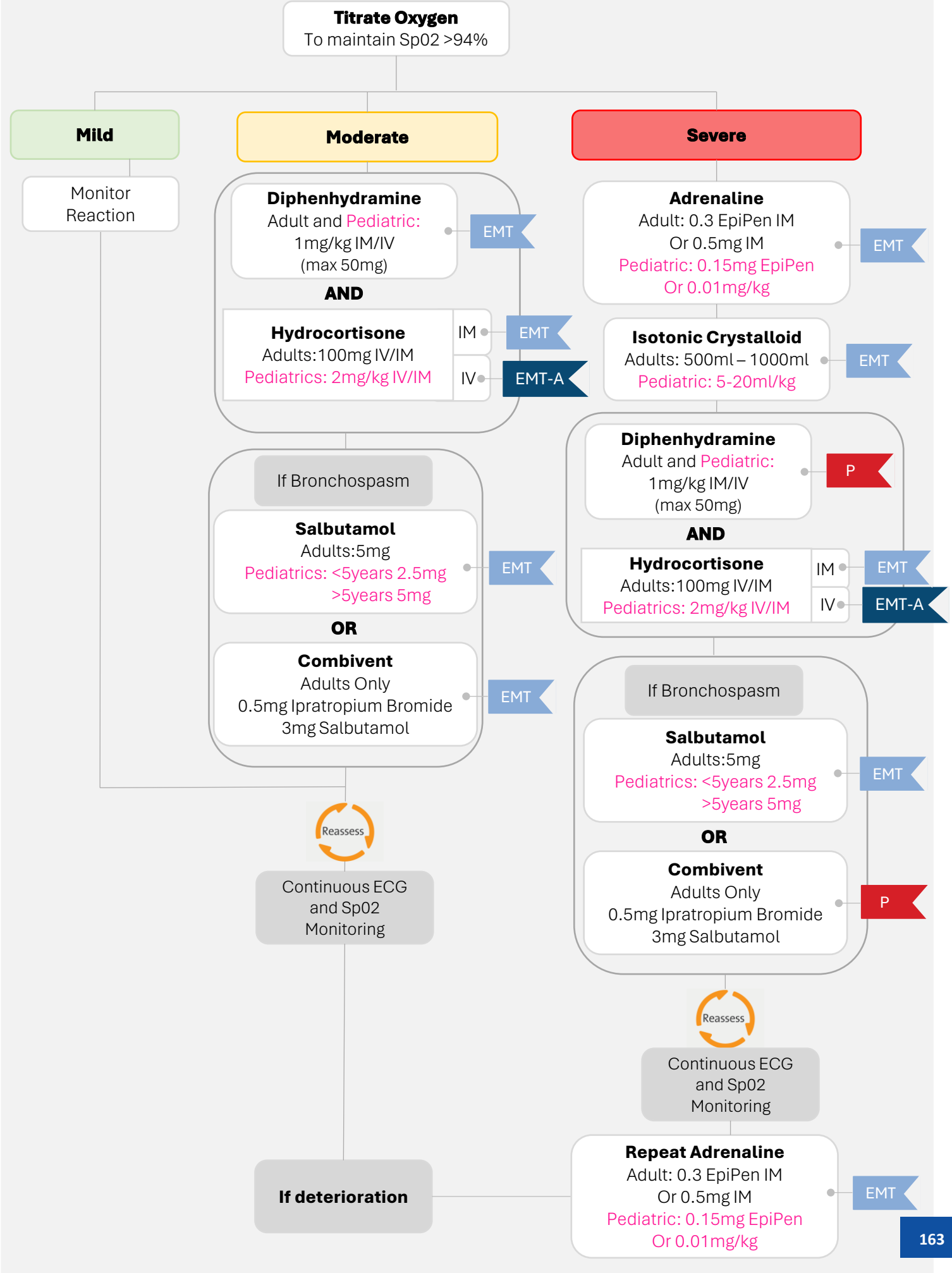
Anaphylaxis / Allergic Reaction



Patient Care Goals

1. Rapidly identify and treat life-threatening anaphylaxis or severe allergic reactions
2. Alleviate symptoms and prevent progression of allergic reactions through timely administration of medications.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with signs and symptoms of an allergic reaction or anaphylaxis, including:
  - Mild to moderate allergic reaction: Localized urticaria (hives), itching, or mild swelling without systemic symptoms.
  - Anaphylaxis: Systemic symptoms involving two or more organ systems (e.g., skin/mucosa, respiratory, cardiovascular, gastrointestinal), such as:
    - Respiratory distress (e.g., wheezing, stridor, shortness of breath).
    - Hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics).
    - Angioedema (swelling of face, lips, tongue, or throat).
    - Gastrointestinal symptoms (e.g., nausea, vomiting, abdominal pain).
  - Known or suspected exposure to an allergen (e.g., food, medication, insect sting, environmental trigger).
  - History of previous allergic reactions or anaphylaxis.

Exclusion Criteria

- None Noted

Key Patient Assessment Points

1. **Allergen Exposure History**
   - Identify recent exposure to potential allergens (e.g., foods, medications, insect stings, latex, or environmental triggers).
   - Confirm history of known allergies or previous anaphylactic episodes.
   - Check for medical alert bracelets or documentation indicating allergy history.
2. **Severity Assessment**
   - Differentiate mild/moderate allergic reaction (e.g., localized hives) from anaphylaxis (systemic involvement of ≥2 organ systems or hypotension).
   - Check for signs of:
     - Airway compromise (e.g., throat swelling, stridor)
     - Respiratory Distress (e.g. bronchoconstriction, wheezing)
     - Shock (e.g., altered mental status, poor perfusion).
3. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary and Secondary Surveys**
   - Perform primary and secondary survey as per Universal Care CPG.
2. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
3. **Consider Red Flags**
   - Rapidly progressing respiratory distress or airway obstruction (e.g., stridor, wheezing, or cyanosis).
   - Severe hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics).
   - Altered mental status or unresponsiveness.
   - Persistent or worsening angioedema, especially involving the airway.
   - Known history of severe anaphylaxis
4. **Evaluate for Signs of Shock or Instability**
   - Hypotension, tachycardia, or poor perfusion (e.g., cool, clammy skin, delayed capillary refill >2 seconds).
   - If anaphylactic shock is present, prioritize Epinephrine administration and fluid resuscitation.
5. **Establish IV Access**

- Indicated for hypotension, moderate to severe symptoms, or anticipated medication administration.
- Consider intraosseous (IO) access if IV access cannot be obtained

6. **Fluid Therapy:**
   - Administer Normal Saline or Ringers Lactate for signs of shock or dehydration:
     - Adults: 500-1000 mL bolus IV, reassess, repeat as needed.
     - Pediatrics: 20 mL/kg bolus IV, reassess, repeat as needed.
   - Monitor for fluid overload, especially in patients with comorbidities.
7. **Mild Reaction**
   - Mild reactions do not routinely require treatment.
   - Generally, a recommendation for transport should be given and the patient reassessed en route to hospital.
8. **Moderate Reaction**
   - **Consider Oxygen**
     - Titrate oxygen to maintain SpO2 ≥94% if hypoxia or respiratory distress is present.
   - **Diphenhydramine**
     - Indicated for urticaria, itching, or mild allergic symptoms.
     - Adults: 25-50 mg IV/IM/PO for mild to moderate allergic reactions or as an adjunct in anaphylaxis.
     - Pediatrics: 1 mg/kg IV/IM/PO (max 50 mg).
   - **Hydrocortisone**
     - Adults: 100-200 mg IV/IM
     - Pediatrics: 2 mg/kg IV/IM (max 200 mg).
     - Note: Administer Hydrocortisone after Epinephrine in anaphylaxis cases.
   - **If Bronchospasm is present:**
     - **Salbutamol**
       - Adults: 2.5-5 mg nebulized for wheezing or bronchospasm.
       - Pediatrics: 2.5 mg nebulized (for <20 kg) or 5 mg (for ≥20 kg).
       - Doses should be diluted with normal saline or sterile water up to a maximum volume of 5ml.
     - **Combivent**
       - Adults: 0.5mg Ipratropium Bromide and 3mg Salbutamol (1nebule)
       - Pediatrics: Not recommended for pediatrics.
       - Doses should be diluted with normal saline or sterile water up to a maximum volume of 5ml.
9. **Severe Reaction**
   - **Epinephrine**
     - Adults: 0.3-0.5 mg IM (1:1000) in the anterolateral thigh, repeat every 5-15 minutes as needed for persistent symptoms.
       - The Adult Epipen is preferred to enable rapid administration
     - Pediatrics: 0.01 mg/kg IM (1:1000, max 0.3 mg), repeat every 5-15 minutes as needed.
       - The Pediatric Epipen is preferred to enable rapid administration
   - **Additional treatment as per Moderate Reaction**
10. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Elderly Patients
  - Anaphylaxis may present atypically (e.g., hypotension without skin symptoms).
  - Use caution with Epinephrine in patients with known cardiovascular disease; consult Telemedicine if needed.

Clinical Insights & Additional Guidance

- Airway Edema
  - Clinicians should have a low threshold for the administration of IM adrenaline in patients with airway edema.
  - Slight increases in airway edema may have a significant impact on airway patency.

- Invasive airway management may be required if symptoms do not reduce – rapid transfer to hospital is required.
- Ensure Dispatch 2 is informed of any patients with significant airway edema or who do not improve to adrenaline.

Documentation

- Identification: Patient's full name, age, and gender.
- Mechanism / Medical Complaint: Exposure to allergen (e.g., food, insect sting, drug), symptom onset and progression.
- Injuries / Information: History of allergies or anaphylaxis, medical alert tags, medications taken before EMS arrival.
- Signs: Vital signs, airway status, SpO_2, skin signs (e.g., rash, angioedema), respiratory gastrointestinal or cardiovascular symptoms.
- Treatment:
    - Epinephrine administration: dose, route, time, number of doses.
    - Other medications given: diphenhydramine, hydrocortisone, salbutamol (dose, route, and time).
    - IV fluids: type, volume, and response.
    - Oxygen therapy details.
- Allergies: Document all known allergies.
- Medications: Pre-existing medications or recent exposures (e.g., NSAIDs, antibiotics).
- Background: Previous history of anaphylaxis, asthma, or cardiovascular conditions.
- Other:
    - Symptom scale (0–10) if used.
    - Communication with Dispatch 2 or Telemedicine.
    - Scene observations or triggers found.

Key Performance Indicators

1. Administer adrenaline for anaphylaxis within 5 minutes of recognition.

References

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